

AI in Healthcare & Banking/Finance

AI in Healthcare

Healthcare is one of the highest-impact domains for AI deployment. AI systems now assist at virtually every stage of care delivery — from diagnostics to drug discovery.

Diagnostics: AI models trained on medical images detect cancer, diabetic retinopathy, and fractures with radiologist-level accuracy.

Drug Discovery: AI (e.g., AlphaFold) predicts protein structures and screens billions of molecular candidates, cutting discovery timelines from decades to years.

ICU Prediction: Early-warning systems flag patient deterioration hours before clinical signs appear, enabling preventive intervention.

Wearables: Smartwatches detect atrial fibrillation, sleep apnoea, and blood oxygen anomalies in real time.

AI in Banking & Finance

Fraud Detection: ML models analyse thousands of transaction features in milliseconds, flagging anomalies with far greater accuracy than rule-based systems.

Algorithmic Trading: AI executes trades in microseconds based on market signals — high-frequency trading now accounts for over 50% of US equity volume.

Risk Simulation: Monte Carlo and ML simulations model portfolio risk, stress-test scenarios, and regulatory capital requirements.

Banking Chatbots: NLP-powered virtual assistants handle millions of routine customer queries daily — balance checks, transfers, loan applications.

AI in finance is characterised by speed, scale, and stakes — a biased or faulty model can move markets or deny credit to millions.

Key Ethical Considerations

- Healthcare: AI misdiagnosis liability; patient data privacy (HIPAA, GDPR); algorithmic bias in clinical trials historically excluded minorities.
- Finance: Black-box credit scoring denying loans; flash crashes caused by correlated algorithmic strategies; regulatory lag.

Key Takeaways

- Healthcare AI: diagnostics, drug discovery, ICU prediction, wearables.
- Finance AI: fraud detection, algorithmic trading, risk modelling, chatbots.
- Both domains demand rigorous accountability, transparency, and regulatory oversight.

Quick-Revision Questions

- Name two ways AI is used in diagnostics and two in drug discovery.
- How does AI-based fraud detection differ from rule-based systems?
- What ethical risks are present in AI-driven credit scoring?